Group 5
Carolina Varanko

Project Progress Log #1

1. **What is your understanding on this project?**

   Our project aims to predict net revenue of top performing Steam games using BI and predictive analytics. Our group will analyze how facts such as price, review score, publisher class, and average playtime influence a game's financial success. The project will produce regression models, a Random Forest analysis, and BI deliverables including a dashboard framed around actionable recommendations for game developers, publishers, and platform operators.

2. **What is your role in this project? Why did you pick this role?**

   My roles in this project include:

   - Title Page/Table of Contents (report)
     - I am responsible for formatting and organizing the report's opening sections to ensure a professional and navigable final document.
   - Dimension 1: High Price vs Low Price tier (e.g. Free-to-Play $0 vs under $20 vs $20-$40 vs +$40)
     - I picked this dimension because pricing is one of the most direct and controllable decisions a developer makes before launch.
   - Regression Analysis (report) (model results/visualizations, presentation)
     - I picked this because of my interest in transforming model findings into clear insights. I will interpret the regression output and build the supporting visualizations for both the report and presentation.
   - Recommendations (presentation)
     - I picked this because I enjoy taking what I have found and turning it into practical takeaways. I will summarize what the data tells us about which factors strongly drive Steam game revenue and translate those results into actionable recommendations for developers and publishers.

3. **What are the data sources?**

   Our primary data source is the *Top 1500 Games on Steam by Revenue* dataset, on Kaggle at https://www.kaggle.com/datasets/alicemtopcu/top-1500-games-on-steam-by-revenue-09-09-2024. The dataset contains 1,500 records and 11 columns including game name, release date, copies sold, price, revenue, average playtime, review score, publisher class, publishers, developers, and Steam ID.

4. **Have you done any data cleaning, and why did you clear the data in such way? E.g., Did you remove any missing value? How many?**

   Data cleaning is being handled by Ethan Rickey as part of our team's division of responsibilities.

   "No, loading the dataset into R, no rows were dropped. Although 1 row was missing data for the publishers column and 2 rows were missing data for developers column, they were left in as our model and analysis does not require those metrics."

5. **Have you done any data reformatting / data transformation data? E.g., How many variables are highly skewed? List the variable name(s)? Did you do any transformation? Why did you choose that particular transformation?**

   Data reformatting/transformation is being handled by Ethan Rickey as part of our team's division of responsibilities.

   "Yes, data transformations was done on all numeric variables used. After loading the dataset, skewness was assessed on all five numeric variables using the moments library in R, which found that all five variables were skewed. Specifically, copiesSold (skewness = 18.60), revenue (22.90), and avgPlaytime (7.16) exhibited substantial positive skewness, price (1.58) showed moderate positive skewness, and reviewScore (−2.05) showed substantial negative skewness. Following the guidelines from Tabachnick and Fidell (2007) and Howell (2007), a Log10 transformation (log10(x)) was applied to copiesSold and revenue, since both are substantially positively skewed with no zero values. For avgPlaytime, a Log10 with constant transformation (log10(x + 1)) was used as it contains one zero value, making a direct log10(0) undefined. For price, a square-root transformation (sqrt(x)) was applied as a milder correction appropriate for moderate skewness, and it handles the 85 free-to-play games, since sqrt(0) = 0 is valid. For reviewScore, which is substantially negatively skewed, a reflect + Log10 transformation (log10(K − x), where K = 101) was applied. Using mutate(), the following new columns were created, each one representing one transformation: log_copiesSold, log_revenue, log_avgPlaytime, sqrt_price, and log_reviewScore. After the transformations, skewness dropped substantially across all variables, copiesSold to 0.98, revenue to 1.18, avgPlaytime to 0.68, price to −0.16, and reviewScore to −0.62, confirming that the transformations were effective in dealing with skewed data."

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6. **What is your time line for the rest of the quarter?**

Now - May 4th: Data cleaning and transformation completed.

May 5th - 11th: Begin setting up price tier dimension analysis using cleaned data.

May 12th - 25th: Individual dimension analyses completed; regression model and Random Forest analysis built and results interpreted.

May 26th - June 1st: BI dashboard and visualizations built; report sections written and finalized.

June 2nd – 8th: All project deliverables completed, PowerPoint presentation complete, group review and final polish; presentation practice.